CrunchBite — Kitchen & Operations Team: Domain Context

Team Scope

The Kitchen & Operations Team owns the **in-store operational experience** — the Kitchen Display System (KDS), speed-of-service (SOS) monitoring, store manager tools, franchise dashboards, and the drive-through workflow. The team also owns the **Store Config Service** (store hours, delivery radius, feature flags per store) and the Medallia NPS integration.

Team Personas

Persona	Description	Key Needs
Kitchen Staff	Prepares orders at individual stations (grill, fry, sandwich, beverage)	Clear order ticket, modifier visibility, easy bump/recall, station-specific routing
Expediter	Assembles complete orders from individual station outputs	Consolidated view of all stations, order-ready indicator, drive-through priority
Drive-Through Attendant	Takes orders and hands them off at the window	Queue time, order status, SOS timer, upsell prompts on OCU
Store Manager	Manages one corporate store	86-ing items, void/refund controls, SOS dashboard, shift management, end-of-day reports
Franchise Owner	Owns 1–15 franchise stores	Multi-store performance roll-up, labour vs target, food cost, SOS trends
Franchise Business Consultant (FBC)	CrunchBite field rep for a territory of 15–30 franchise stores	Territory-level SOS and compliance dashboards, franchisee alert management

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Kitchen Display System (KDS)

Hardware

- **Epson KDS hardware** running the CrunchBite KDS app
- Mounted at each kitchen station (grill, fry, sandwich, beverage) and at the expediter position
- A separate **manager/drive-through screen** displays SOS timers and queue status

Ticket Workflow

- Every new order creates a **KDS ticket** showing: order ID, order type (dine-in / pickup / delivery / drive-through), items grouped by station, modifiers, and elapsed time
- **Station routing**: ticket is delivered to the relevant station screen(s) simultaneously
- **Make time targets**: 3 minutes for standard orders; 5 minutes for catering items
- **Bump**: kitchen staff taps an item when it is ready; the ticket clears from the station screen when all items at that station are bumped

- **Recall**: last 20 bumped tickets accessible for re-display (e.g., if an order is lost or incorrect)
- **Expediter screen**: consolidates all station outputs; shows a complete order as assembled when all stations have bumped; expediter bumps the assembled order to indicate it is ready for handoff

Drive-Through Specific

- **SOS Timer**: order-placed → order-ready timer shown on the drive-through attendant screen
- Transaction time target: **3 minutes 30 seconds** order-to-window
- 240 of the 680 corporate stores have dual-lane drive-through; each lane shown as a separate queue on the expediter screen
- **Order Confirmation Unit (OCU)**: customer-facing display at the order point; shows cart in real time; AI suggestive selling prompt shown based on order content and daypart

AI Suggestive Selling (Drive-Through)

- Upsell prompt appears on the OCU after the initial order is placed
- Prompt is generated by a rules engine (not ML in Phase 1) based on: current order content, daypart, and top-selling add-ons for that store
- Phase 2: ML-driven personalised upsell based on LPR (licence plate recognition) linked to loyalty history

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86-ing (Item Unavailability) — Store Manager View

- Store manager marks an item unavailable via the **Manager Dashboard** (tablet or desktop)
- The action publishes an `item.86d` Kafka event consumed by the Menu Service, which propagates to all ordering channels within 60 seconds
- Managers can also view which items are currently 86'd at their store and restore them individually
- Corporate can 86 an item across all stores simultaneously (used for supply chain issues)

Speed-of-Service (SOS)

- SOS is measured as: order placed timestamp → order ready (all items bumped at expediter) timestamp
- Displayed in real time on the manager dashboard per order and as a rolling store average
- **Target**: 3 minutes 30 seconds drive-through; 3 minutes in-store
- SOS breaches (orders exceeding target by more than 30 seconds) are flagged on the manager dashboard
- Store-level SOS metrics feed into the franchise owner roll-up dashboard (via Snowflake/Tableau)

Store Manager Controls

- **Void/refund**: managers may void orders up to $75 without approval; above $75 requires a district manager PIN
- **Store configuration**: store hours, delivery radius, per-store feature flags — managed via the Store Config Service
- **86-ing**: real-time item unavailability marking with 60-second propagation SLA
- **End-of-day report**: summary of orders, gross sales, voids, SOS average, and items 86'd during the shift

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Franchise & Multi-Store Dashboards

- **Franchise Owner view**: multi-store roll-up of sales, SOS performance, top-selling items, labour cost vs target, food cost percentage; updated every 15 minutes (Snowflake)
- **FBC view**: territory-level aggregation across 15–30 stores; includes compliance scoring (e.g., % of stores meeting SOS target) and franchisee alert queue
- **KPIs tracked**: GTV (gross transaction value), average ticket size, SOS average, table turns, drive-through throughput, NPS score

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NPS & Guest Feedback

- **Medallia** NPS survey triggered automatically after every completed order (post-transaction webhook from Order Service)
- Survey delivered via email or SMS (based on member preferences); link-based for guest orders
- NPS responses and verbatim feedback flow into the franchise dashboard and Snowflake for trend analysis
- Store managers see their store's NPS score and verbatim comments in the Manager Dashboard

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Business Rules Specific to Kitchen & Operations

1. KDS ticket must appear on the correct station screen(s) within **1 second** of the order being placed
2. A store manager can only 86 items at **their own store**; corporate and FBCs have broader scope
3. Voids above $75 require a **district manager PIN** entered at the POS terminal (not the digital dashboard)
4. SOS timer starts the moment the `order.placed` Kafka event is consumed by the Kitchen Service
5. Dual-lane drive-through stores display **two separate queue views** on the expediter screen
6. Catering orders appear **only on the expediter screen** — not on individual station bump screens — to avoid confusion during normal service